

Fig. 2: Top three growth drivers of the PCB industry

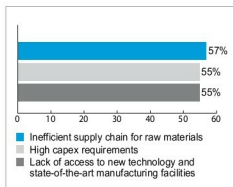


Fig. 3: Top three challenges hampering the growth of PCB manufacturing in India

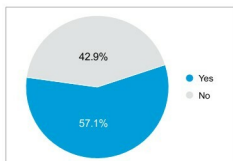


Fig. 4: Will the industry grow inorganically?

participants have also unanimously stated that the industry is poised for growth during FY 2017-18. However, there are mixed opinions about the percentage of expected growth (Fig. 1).

Growth drivers

Considering the estimated demand till 2020 (Table I), growth of the PCB market will most likely be driven by the consumer electronics segment, followed by requirements from industrial and LED lighting segments.

Survey participants, however, feel that demand scenario in 2017-

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18 may differ slightly from the forecast indicated in Table II. According to them, the automotive industry, LED lighting and industrial electronics will be the major growth drivers (Fig. 2). Mobile phones, consumer electronics devices and broadcast and telecommunication equipment will also contribute significantly in demand generation.

Miniaturisation of semiconductor devices and enhanced functionality in electronic products is expected to drive the demand for more complex PCBs with a higher number of layers, in large volumes. PCB manufacturers in the country mostly fall into the following segments:

1. Large-volume manufacturers (primarily double-sided PCB makers and a few who produce a good amount of multi-layered PCBs)
2. Small-volume manufacturers (mostly double-sided PCB makers and a few who produce multi-layered PCBs)
3. Small-volume fast-delivery manufacturers (mostly double-sided PCB makers and a few who produce multi-layered PCBs)
4. Small-volume complex-design PCB manufacturers

The survey participants expect growth in the second, third and fourth categories.

Some of the survey participants have indicated the possibility of demand arising out of advances in technology such as high-density interconnection. This enables more interconnection functions per unit area, which will positively impact the market for advanced-level applications. In addition, the recent

thrust towards indigenisation in strategic electronics manufacturing is likely to fuel the demand for high-grade, multi-layered PCBs.

Challenges faced by the industry

Currently, output of the Indian PCB industry is less than one per cent of the total global capacity. According to survey participants, top challenges in the country face are:

1. Inefficient supply chain for raw materials
2. High capex requirements
3. Lack of access to new technology and state-of-the-art manufacturing facilities

Other challenges include:

1. Unfair playing field, since companies from competing countries have access to finance at much lower costs
2. Logistics inefficiencies and infrastructural bottlenecks, resulting in longer turnaround time frames and higher costs
3. Conflicting policies in operation across various levels within the electronics industry, from the components segment to finished products

An inefficient supply chain and non-availability of a significant percentage of critical raw materials increase inventory costs. Capital costs also go up due to the very high cost of finance in India—both for capital investments as well as for working capital. The very low capital-output ratio in the PCB industry entails the need for huge investments and high interest costs. Thus, profitability is